IMPERIAL COLLEGE LONDON

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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected copy

JOHN MCCLENN 2/5/19

Seema · 2/5/19

OPTIMIZATION

Thursday, 2 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : P.L. Dragotti

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OPTIMISATION

1. Consider the function

$$f(x_1,x_2) = \frac{(x_1+x_2-2)^2}{2} + (r-1)\frac{(x_1-x_2)^2}{2},$$

where $r \geq 2$ is an integer.

a) Compute the stationary point $X_\star$ of the function f. Show that the stationary point is a global minimizer for f.
(Hint: recall that r is an integer not smaller than 2.) [4 marks]

b) Write the gradient iteration for the minimization of f. [2 marks]

(Hint: to avoid confusion, you may wish to use $\{X_k\}$ to indicate the sequence generated by the gradient algorithm, with $X_k = (X_{k,1}, X_{k,2})$ the k-th element of the sequence).

c) Consider the gradient iteration in part b) and assume $r = 100$.

i) Consider the initial point $X_0 = (X_{0,1}, X_{0,2})$ such that $X_{0,1} = 5$. Determine the second component of X_0, that is $X_{0,2}$, such that the gradient of f at X_0 is parallel to the x_1 axis. [4 marks]

ii) Run one iteration of the gradient method with exact line search, that is determine the point X_1. Show that the exact line search parameter is $\alpha_0^\star = 1/100$. Show that the gradient of the function f at X_1 is parallel to the x_2 axis. [4 marks]

iii) Run, again, one iteration of the gradient method with exact line search, that is determine the point X_2. Show that, again, the exact line search parameter is $\alpha_1^\star = 1/100$. Show that the gradient of the function f at X_2 is parallel to the x_1 axis. [4 marks]

iv) Note that the gradient iteration generates a sequence $\{X_k\}$ such that the sub-sequence $X_{\text{even}} = \{X_0, X_2, X_4, \cdots, X_{2k}, \cdots\}$ lays on a straight line connecting the initial point X_0 to the global minimizer $X_\star$. Assume that, on average, the norm of the gradient of f restricted to this straight line is equal to 1. Hence determine the average distance between X_{2k} and X_{2k+2} and estimate the number of iterations required to reach the point $X_\star$. [2 marks]

(Note: as a matter of fact, the sequence X_{even} converges to $X_\star$ asymptotically. The above estimate is only useful to reveal that the gradient method is very slow in the considered scenario. This is a well-known scenario for the gradient method (and also other methods) and it is often referred to stating that the algorithm is caught in the cage of Stiefel!)

2. Consider an optimization problem with objective function

$$f(x,y) = x + 2y$$

and decision variables x and y such that the constraint

$$g(x,y) = y^2 + xy - c^2 = 0$$

holds. Assume $c > 0$.

a) Write the necessary conditions of optimality for this problem and determine candidate optimal solutions as a function of the constant c. [2 marks]

b) Using second order sufficient conditions of optimality classify the candidate optimal solutions determined in part a), that is determine which one is a local (strict) minimizer and which one is a local (strict) maximizer. [4 marks]

c) Compute the value of the objective function at the local minimizer and at the local maximizer determined in part b). Show that the value of the objective function evaluated at the local maximizer is smaller than the value of the objective function evaluated at the local minimizer. Explain which conditions might give rise to this situation. [6 marks]

d) Show that the (total) derivative with respect to the parameter c of the objective function evaluated at the local strict minimizer equals the partial derivative with respect to c of the Lagrangian evaluated at the optimal multiplier. [2 marks]

e) The considered problem can be transformed into an unconstrained problem by solving the constraint in the variable x, that is by eliminating the variable x, yielding an objective function $f(x(y),y)$ which depends only upon the variable y. Compute explicitly the objective function $f(x(y),y)$ and explain, drawing the graph of the function, why the local minimizer/maximizer does not actually solve the given minimization/maximization problem. [4 marks]

f) Assume now that $c = 0$. Show, using first order necessary conditions of optimality that there are no candidate optimal solutions. Consider the limit as $c \to 0$ of the function $f(x(y),y)$ computed in part e). Use the properties of this function to explain why there are no candidate optimal solutions. [2 marks]

is it referring to $f(x,y)$ or $f(x(y),y)$?

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3. A multi-objective constrained optimization problem with s objectives and p inequality constraints can be formulated as follows:

$$\min_x f(x),$$
$$h(x) \leq 0,$$

with $x \in \mathbb{R}^n$, $f : \mathbb{R}^n \to \mathbb{R}^s$ and $h : \mathbb{R}^n \to \mathbb{R}^p$. Note that there are s objective functions. This problem may have multiple (sometimes even infinitely many) solutions depending on the notion of optimality. In particular a feasible point $x_\star$ is called Pareto optimal, or a Pareto point, if there does not exist another feasible point $\tilde{x}$ such that

(C1): $f_i(\tilde{x}) \leq f_i(x_\star)$, for all $i \in [1, \cdots, s]$;
(C2): $f_k(\tilde{x}) < f_k(x_\star)$, for at least one $k \in [1, \cdots, s]$.

In other words, a feasible point is Pareto optimal if no objective can be improved without penalizing another objective.

Consider now a multi-objective constrained optimization problem with $n = 2$, $s = 2$, $p = 3$ and

$$f(x) = \begin{bmatrix} f_1(x_1,x_2) \\ f_2(x_1,x_2) \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad h(x) = \begin{bmatrix} 4 - x_1^2 - x_2^2 \\ x_2 - 2x_1 \\ x_1 - 2x_2 \end{bmatrix}.$$

a) Plot the admissible set and the level lines of the objective functions f_1 and f_2. [2 marks]

b) Using the definition of Pareto optimality and the plot from part a) determine all Pareto optimal solutions. Explain why the Pareto optimal points are not unique and why one often refers to such points as South-West points. [4 marks]

c) The considered multi-objective constrained optimization problem can be solved using the so-called weighted problem, that is the problem

$$\min_{x_1,x_2} w_1 x_1 + w_2 x_2,$$
$$4 - x_1^2 - x_2^2 \leq 0,$$
$$x_2 - 2x_1 \leq 0,$$
$$x_1 - 2x_2 \leq 0,$$

where w_1 and w_2 are strictly positive constants.

i) State first order necessary conditions of optimality for the weighted problem. [4 marks]

ii) Use the necessary condition of optimality in part c.i) to determine conditions on w_1 and w_2 guaranteeing that the candidate optimal solutions of the weighted problems recover some, or all, of Pareto optimal solutions. [10 marks]

4. Consider the optimization problem $\min_x f_1(x) + f_2(x)$, with $x \in \mathbb{R}^n$.

This problem can be re-formulated as the constrained optimization problem, referred to as a distributed optimization problem,

$$\min_{x_1, x_2, z} f_1(x_1) + f_2(x_2),$$
$$x_1 - z = 0, \qquad x_2 - z = 0,$$

with $x_1 \in \mathbb{R}^n$, $x_2 \in \mathbb{R}^n$, and $z \in \mathbb{R}^n$.

a) Show that any candidate optimal solution of the original problem yields a candidate optimal solution of the associated distributed optimization problem, and vice versa. [4 marks]

b) The distributed optimization problem can be solved using the distributed algorithm

$$x_1^{k+1} = \text{argmin}_{x_1} \left(f_1(x_1) + (\lambda_1^k)'(x_1 - z^k) + \frac{\eta}{2}\|x_1 - z^k\|^2 \right),$$

$$x_2^{k+1} = \text{argmin}_{x_2} \left(f_2(x_2) + (\lambda_2^k)'(x_2 - z^k) + \frac{\eta}{2}\|x_2 - z^k\|^2 \right),$$

$$z^{k+1} = \frac{x_1^{k+1} + x_2^{k+1}}{2} + \frac{\lambda_1^k + \lambda_2^k}{2\eta},$$

$$\lambda_1^{k+1} = \lambda_1^k + \eta(x_1^{k+1} - z^{k+1}),$$

$$\lambda_2^{k+1} = \lambda_2^k + \eta(x_2^{k+1} - z^{k+1}),$$

where $\text{argmin}_\xi F(\xi, \kappa)$ has to be understood as the operation of finding the minimizer of the function F with respect to the variable ξ assuming κ is a parameter, and $\eta > 0$ is a penalty term.

i) Let $x \in \mathbb{R}$ and

$$f_1(x) = \frac{1}{2}(x-1)^2, \qquad f_2(x) = \frac{1}{2}(x+2)^2.$$

Find the unique solution of the considered optimization problem.

[2 marks]

ii) Write the distributed optimization algorithm for this problem. Note that the argmin operation can be explicitly computed, since the functions to be minimized are quadratic and convex. [4 marks]

Show that if $\lambda_1^k + \lambda_2^k = 0$, for all $k \geq 0$, then the equation describing z^{k+1} does not depend upon x_1^k, x_2^k, λ_1^k and λ_2^k. Under these conditions determine the value $z^\infty = \lim_{k\to\infty} z^k$ and show that this is the same as the optimal point determined in part b.i). [4 marks]

Let $\eta = 100$, $x_1^1 = 1$, $x_2^0 = -2$, $z^0 = 0$, $\lambda_1^0 = \lambda_2^0 = 0$. Run two iterations of the distributed optimization algorithm, that is determine x_1^2, x_2^2, z^2, λ_1^2 and λ_2^2, and argue that the condition $x_1 \approx x_2$ is achieved after two steps. Note also that the condition $\lambda_1^k + \lambda_2^k = 0$, for all $k \geq 0$, implies that z is an average of x_1 and x_2 and, hence, describe how convergence of x_1 and x_2 to the optimal solution determined in part b.i) is achieved. [6 marks]